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A unified model for species fluxes, ion impact energy and heat transfer at the electrode of an atmospheric-pressure arc discharge

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Abstract

1 Introduction

Arc discharges find extensive applications in welding, cutting, and thermal spraying. In recent years, their use has expanded to emerging applications, including plasma pyrolysis, advanced material synthesis, and metallurgical reduction processes [1, 2, 3]. An atmospheric-pressure arc discharge spans a wide range of length scales, differing by several orders of magnitude. The millimeter-to-centimeter-scale quasi-neutral arc column constitutes the bulk of the discharge, where convection, thermal conduction, Joule heating, and radiation interact under conditions that are often close to local thermodynamic equilibrium (LTE). In contrast, the sub-micron to sub-millimeter regions adjacent to the electrodes are characterized by strong nonequilibrium effects. In these regions, the plasma is subjected to the relatively cold electrode surfaces, steep density gradients associated with particle absorption at the electrodes, and strong electric fields arising from charge separation. These fields can accelerate electrons and ions to energies substantially higher than their local thermal energies.

Resolving the full range of length scales throughout the entire discharge is neither computationally practical nor necessary for many applications. Consequently, arc-discharge models commonly treat the quasi-neutral bulk plasma separately from the nonequilibrium electrode regions. In the bulk, the quasineutrality condition, $n_i \approx n_e$, eliminates the need to explicitly solve Poisson's equation, while the near-LTE assumption further reduces model complexity by allowing thermodynamic and transport properties to be determined primarily from the local temperature and pressure. Macroscopic effects such as cathode jets, electrode geometry, convection, and electromagnetic forces can then be incorporated within a continuum formulation based on the Navier–Stokes equations with Joule-heating and Lorentz-force terms.

Quasi-neutral bulk plasma modeling, can be broadly classified into four categories according to the degree of thermodynamic and chemical equilibrium assumed and the treatment of electron and heavy-particle temperatures.

****(i) Full local thermodynamic equilibrium.**** Experimental measurements of species temperatures and electron densities indicate that the arc core is generally close to LTE [4, 5, 6]. In a full LTE model, electrons and heavy particles are characterized by a single temperature T, with Maxwellian velocity distributions, Boltzmann excitation, Saha ionization, and Dalton's law used to determine the plasma state and properties. The governing equations typically consist of mass, momentum, energy, and current continuity [7, 8, 9]. Conventional LTE models terminate before the electrode sheaths, where the assumptions of LTE and quasineutrality break down. Moreover, as $n_e \to 0$ near an absorbing electrode, $\sigma \to 0$, making the current-continuity equation, $\nabla \cdot (\sigma \nabla \phi) = 0$, increasingly ill-conditioned as the mesh is refined. Lowke and Tanaka's LTE-diffusion approximation [10] avoids this difficulty by limiting the near-electrode cell size to the electron diffusion length, $\Lambda = D_e/W = (kT_e/e)/E_z$, while retaining LTE properties. This approach reproduces the arc voltage and electrode heat fluxes to within approximately 10% of a full nonequilibrium treatment.

****(ii) Two-temperature and two-temperature-equilibrium models.**** Two-temperature (2T) models treat electrons and heavy particles as separate Maxwellian populations at T_e and T_h,

coupled through elastic energy exchange. In two-temperature-equilibrium (2T-EQ) models, the chemical composition is additionally determined from a generalized Saha relation for $T_e \neq T_h$. Hsu and Pfender [11] showed that $T_e \approx T_h$ in the high-density arc core, with significant temperature separation emerging in the low-density fringes and near the anode. However, the generalized Saha closure is not unique: different formulations can predict electron densities differing by up to an order of magnitude away from $T_e/T_h \approx 1$ [12]. Furthermore, inconsistencies in the partitioning of ionization-energy and reactive thermal-conductivity terms between the electron and heavy-particle energy equations have been identified in several formulations [13].

****(iii) Nonequilibrium models.**** Fully nonequilibrium models abandon equilibrium closures such as LTE and generalized Saha equilibrium, allowing the plasma composition and electron and heavy-particle temperatures to evolve independently. These models typically include multicomponent species transport and separate electron and heavy-particle energy equations. Examples include the nonequilibrium arc models of Haidar [14], Gonzalez *et al.* [15], and Amakawa *et al.* [16]. By resolving departures from equilibrium directly, these models can describe the transition from the LTE-like bulk plasma toward the strongly nonequilibrium electrode region. However, the space-charge sheath is generally still treated separately using an appropriate sheath model.

****(iv) LTE with diffusion-driven ionization nonequilibrium.**** Another approach considered in the literature retains $T_e = T_h = T$ while allowing the electron and ion densities to depart from their local Saha-equilibrium values. This departure arises when diffusion competes with the ionization–recombination relaxation processes, particularly near the electrodes. Lowke [10] demonstrated this approach using a full-diffusion calculation that solves the ambipolar electron-continuity equation together with current continuity, including the electron-diffusion contribution. The resulting solution produces a retarding-field potential structure near the electrode that is absent in a conventional LTE calculation. Tanaka and Lowke [17] further incorporated Poisson's equation to account for space-charge effects and reported relatively small sheath voltage drops compared with the total arc voltage. However, because $T_e = T_h$ is retained throughout the domain, this formulation cannot capture the strong electron–heavy-particle temperature separation expected in the low-density electrode-adjacent region.

1.1 Coupling the bulk plasma to the electrode: sheath models

Regardless of the bulk-plasma model employed, the bulk solution must ultimately be coupled to a description of the electrode region. The appropriate coupling strategy depends on the assumptions made in the bulk model and on which nonequilibrium processes are retained.

For full LTE models (i), the electrode region must account for the breakdown of LTE, including the separation between electron and heavy-particle temperatures, deviations from Saha equilibrium, and the formation of a space-charge sheath. One possible approach is *unified modeling*, in which a single set of governing equations, including Poisson's equation, is solved across the entire inter-electrode region without explicitly separating the bulk plasma from the sheath. This approach provides a self-consistent description of the electrode region but is computationally expensive because the mesh must resolve the extremely small sheath length scales. Consequently, applications to arc discharges have largely been restricted to one-dimensional calculations extending from the electrode into the bulk plasma [18, 19].

For two-temperature and two-temperature-equilibrium models (ii), the bulk description already accounts for electron–heavy-particle temperature differences, but deviations from ionization equilibrium and the formation of a space-charge sheath must still be addressed. A common approach is therefore to introduce an *ionization layer*: a thin, quasi-neutral region adjacent to the electrode that connects the bulk plasma to the sheath. This layer provides an intermediate region in which ionization equilibrium breaks down before the plasma enters the strongly nonequilibrium sheath. The sheath is then treated as a separate space-charge region between the ionization layer and the electrode, commonly using a collisionless sheath model.

For fully nonequilibrium bulk models (iii), no additional ionization layer is necessarily required because the bulk formulation already allows departures from equilibrium. The remaining transition to the electrode is associated primarily with the breakdown of quasineutrality and the formation of the space-charge sheath. The sheath can therefore be coupled directly to the nonequilibrium quasi-neutral plasma using an appropriate sheath model. However, the validity of simplified analytical sheath models and their matching conditions depends on the collisionality of the sheath. Benilov *et al.* [20] noted that direct matching to an analytical sheath solution is strictly justified only under the corresponding assumptions regarding sheath collisionality.

The limitations of the commonly adopted collisionless-sheath approximation were examined

directly by Almeida, Benilov, and Naidis [18], who evaluated the number of ion collisions occurring as an ion traverses the cathode space-charge sheath, $\int \left(\lambda_{ia}^{-1} + \frac{\bar{\nu}_{ia}}{|v_i|}\right) dx$. For mercury and atmospheric-pressure argon arcs at practically relevant current densities, they obtained values ranging from approximately 10^2 to 10^4. Thus, ions can undergo a substantial number of collisions while crossing the sheath, demonstrating that the assumption of collision-free ion motion, which underlies many simplified electrode models, is generally not justified at high pressure. Even for the comparatively well-behaved atmospheric-pressure argon case, the ion-collision number is far greater than unity rather than $\ll 1$. This result highlights the need for a self-consistent treatment of collisional ion transport in the near-electrode region.

Except in the immediate vicinity of the electrodes, an atmospheric-pressure arc is sufficiently close to local thermodynamic equilibrium (LTE) that the bulk arc column can be modeled without explicitly resolving the nonequilibrium regions in detail. The LTE-diffusion approximation of [10] demonstrates this explicitly, reproducing the arc voltage and electrode heat flux obtained from a full nonequilibrium treatment to within approximately 10%, while retaining a single-temperature LTE closure throughout the computational domain. The temperature distributions obtained using this approach also show good agreement with experimental measurements.

The approach adopted in the present work therefore retains the computationally efficient LTE treatment for the bulk of the arc while explicitly resolving the nonequilibrium physics in the near-electrode region. Rather than prescribing electrode-boundary quantities or relying on an analytical sheath model, a unified drift-diffusion–Poisson formulation is employed to simultaneously determine the charged-particle distributions, species fluxes, ion impact energies, heat transfer to the electrode, and sheath voltage drop. The same governing drift-diffusion–Poisson system is solved continuously from the quasi-neutral bulk plasma through the near-electrode transition region and into the sheath. The highly disparate spatial scales are resolved using adaptive mesh refinement (AMR), with the computational mesh refined to a fraction of the local Debye length in the sheath region.

An additional advantage of this approach is that it does not require a separate sheath closure based on a prescribed sheath polarity. Field reversals and changes in the sign of the potential drop near the anode have been reported in atmospheric-pressure arcs and can depend on several factors, including the attachment mode, current density, and electrode ablation effects [21, 22, 23]. By solving the drift-diffusion–Poisson system self-consistently across the near-electrode region, the sheath potential structure and its polarity emerge from the coupled solution rather than being imposed a priori. This approach enables one to use the results from the computational efficient bulk plasma treatment to inform the boundary conditions for the near-electrode region, while simultaneously capturing the nonequilibrium effects that arise in the sheath and ionization layer. The model and its numerical implementation in the PlasmeX code are described in Section 4.

2 Physical model and scale analysis

2.1 Governing equations

In the electrode region, the plasma interacts with the relatively cold, absorbing electrode surfaces. This interaction introduces strong gradients in the plasma properties, causing the plasma to progressively depart from LTE. We consider plasma to be composed of electrons, singly-charged ions, and neutral atoms for the model. The presence of multiply charged species is neglected as justified in [24]. To estimate the characteristic length scales associated with these departures, we consider the adjacent LTE plasma at two representative temperatures, $T = 10{,}000$ K and $T = 20{,}000$ K, which are typical of the anode and cathode regions, respectively, in a high-intensity atmospheric-pressure arc discharge [10, 9, 25].

In an ionized gas, the electron mean free path can be approximated as $\lambda_e = 1/(n_e\overline{Q}_{ee} + n_a\overline{Q}_{ea})$, where n_e and n_a are the electron and neutral-atom number densities, respectively, and $\overline{Q}_{ee}$ and $\overline{Q}_{ea}$ are the corresponding momentum-transfer collision cross sections for electron–electron and electron–atom collisions. Electron–electron collisions are Coulomb collisions, for which the averaged momentum-transfer cross section is given by $\overline{Q}_{ee} = e^4 \ln\Lambda/[32\pi\varepsilon_0^2(k_B T_e)^2]$ [26]. Here, $\ln\Lambda$ is the Coulomb logarithm, given by $\ln\Lambda = 1.24 \times 10^7 T_e^{3/2} n_e^{-1/2}$, where T_e is in K and n_e is in m^{-3}.

The average electron–neutral momentum-transfer cross section is of the order 0.2×10^{-19} m^2 [27] for the electron temperatures considered. Using the corresponding LTE electron and neutral densities, the resulting electron mean free path is of order $\lambda_e \sim 10^{-6}$ m in the electrode regions. The characteristic length scale for electron–heavy-particle energy relaxation, λ_T, is approximated as $\frac{\lambda_e}{\sqrt{2M/m_e}} \sim 2 \times 10^{-4}$ m, where M is the mass of the heavy species (argon in this case) and m_e is the electron mass. Thus, $\lambda_T \sim 10^{-4}$ m.Lowke and Tanaka [10] identified a characteristic diffusion

length scale for the electron and ion densities of $\lambda_d \sim 10^{-4}$ m in the electrode regions.

At the smallest spatial scale, the Debye length calculated from the LTE number densities is of order $\lambda_{De} \sim 10^{-8}$ m, which characterizes the spatial extent over which charge separation and space-charge effects become important. The ion mean free path provides another relevant scale and can be estimated as $\lambda_i = 1/(n_a \overline{Q}_{ia})$, where $\overline{Q}_{ia}$ is the momentum-transfer cross section for ion–atom collisions. For argon, we use the approximate fit $\overline{Q}_{ia} = 2[7 \times 10^{-10} - 0.6 \times 10^{-10} \ln(2k_B T_h)]$ [28], where k_B is the Boltzmann constant and T_h is the heavy-particle temperature. Since the heavy particle temperature vary from LTE condtions $\sim$ 1000K to $\sim$ 1000K, to electrode temp and since we are interested in region close to the electrode, we use a temperature of $T_h \sim 1000$ K, which gives an ion mean free path of order $\lambda_i \sim 10^{-7}$ m.

The resulting hierarchy of characteristic length scales in the near-electrode region is therefore $\lambda_{De} \ll \lambda_i \ll \lambda_e \ll \lambda_d \sim \lambda_T$. These estimates indicate that the near-electrode region over which significant departures from LTE develop extends over a characteristic length scale of order 10^{-4} m. Other modeling results results and analysis shows the same [18, 29, 30, 31]. Consistent with this estimate, Lowke and Tanaka [10] showed that excluding a near-electrode region of approximately 10^{-4} m is sufficient to model the remaining plasma using an LTE approximation. This separation of scales provides the basis for treating the bulk arc with an LTE model while explicitly resolving the nonequilibrium physics in the near-electrode region with a 1D model due to the small size of the near-electrode region.

We therefore model the electrode region as a one-dimensional domain, with the electrode surface at one end and the bulk plasma at the other. Within this region, fluid-dynamic effects associated with bulk plasma motion are neglected, since the near-electrode region is typically contained within the hydrodynamic boundary layer adjacent to the electrode. The neutral species are assumed to be stationary and in thermal equilibrium with the electrode surface and with each other. The ions are assumed to remain in thermal equilibrium with the neutrals because their mean free path is small compared with the characteristic length scale of the temperature variation. Accordingly, the heavy-particle temperature is prescribed using an exponential profile that decreases from the bulk plasma temperature to the electrode temperature,

$$T_h(x) = T_{\text{bulk}} + (T_{\text{wall}} - T_{\text{bulk}}) \exp\left(-\frac{x}{L_T}\right),$$

where L_T is a characteristic thermal length scale. This exact ion temperature profile makes a little difference on the model because the coliision frequency if ion-neutral has a weak dependence on collision energy. The electron number density and temperature, and the ion number density, are therefore solved self-consistently, while the heavy-particle temperature is prescribed. The neutral number density is not solved independently but is obtained from the ideal-gas law using the prescribed heavy-particle temperature and the constant system pressure. The system is closed by Poisson's equation for the electric potential, which is solved throughout the entire computational domain.

In addition, inertial effects are neglected, and the frictional, pressure-gradient, and electric-field forces are assumed to balance locally for each charged species. This approximation corresponds to a quasi-steady drift-diffusion description in which the species flux adjusts rapidly to changes in the local plasma conditions. It is valid when the momentum-relaxation time is sufficiently short compared with the characteristic timescale of the macroscopic evolution, or equivalently when the associated momentum-relaxation length is small compared with the characteristic spatial scale of the plasma properties. The validity of this approximation is discussed in Section fill later. The resulting quasi-steady flux is then used in the species continuity equations to determine the spatial evolution of the charged-particle densities.

The continuity equation for species s is given by

$$\frac{\partial n_s}{\partial t} + \nabla \cdot \mathbf{\Gamma}_s = S_s, \qquad \mathbf{\Gamma}_s = z_s \mu_s n_s \mathbf{E} - D_s \nabla n_s, \tag{1}$$

where $z_s = \pm 1$ is the signed charge number, μ_s is the mobility, D_s is the diffusion coefficient, and S_s is the net ionization–recombination source term. For the simple two-body ionization/recombination scheme considered here, the source terms for electrons and ions are equal in magnitude and opposite in sign. The flux $\mathbf{\Gamma}_s$ is the particle flux of species s; its detailed formulation and the calculation of the corresponding transport properties are described in Appendix fill later. The ionization and recombination processes are modeled using the formulation of Hoffert and Lien [32].

The electric field is obtained from Poisson's equation,

$$\nabla \cdot (\varepsilon_0 \mathbf{E}) = e(n_i - n_e), \tag{2}$$

which is solved over the entire computational domain, thereby allowing the transition from the quasi-neutral bulk plasma to the space-charge region to be captured without imposing a priori sheath boundaries.

The electron energy is governed by a separate equation of the same convection–conduction–Joule-heating–energy-exchange form commonly used in two-temperature bulk-arc models [11, 13],

$$\frac{3}{2}\frac{\partial(n_e T_e)}{\partial t} + \nabla\cdot\left(\frac{5}{2}T_e\mathbf{\Gamma}_e - \kappa_e\nabla T_e\right) = \mathbf{E}\cdot\mathbf{J} - \dot{E}_{eh} - \dot{E}_{\mathrm{inel}}, \tag{3}$$

where the terms on the right-hand side represent Joule heating, elastic electron–heavy-particle energy exchange, and inelastic energy losses, respectively. The elastic energy-exchange term is given by $\dot{E}_{eh} = n_e\delta\nu_{en}\frac{3}{2}k_B(T_e - T_{\mathrm{gas}})$, where $\delta = 2m_e/M$, while $\dot{E}_{\mathrm{inel}}$ accounts for energy losses associated with inelastic processes, including ionization. Radiation losses are usually small [29, 18] and thus not included.

Heavy-particle temperature is either taken as a fixed background gas temperature $T_{\mathrm{gas}}(x)$ (the regime in which the present implementation, Section 4, currently operates) or evolved by its own energy equation as in [11, 18] when a fully coupled two-temperature heavy-species treatment is required.

3 Appendix A. Transport Coefficients

The electron mobility is obtained from a two-term Boltzmann solution of the BOLSIG+-type, assuming a Maxwellian electron energy distribution, and is tabulated in reduced form as $\mu_e N(T_e)$ and $D_e N(T_e)$, rather than being evaluated from a closed-form kinetic-theory expression. This treatment is appropriate in the weakly ionized regime, where electron–neutral collisions dominate and the electron energy distribution is close to Maxwellian. The validity of the Maxwellian assumption is discussed in Section fill later. However, in the core region of the arc, particularly near the cathode, electron–ion Coulomb collisions can become significant. To provide a continuous description between these regimes, an electron–ion Coulomb correction is incorporated into the BOLSIG+-based transport coefficients.

The electron–neutral collision frequency is obtained from the BOLSIG+-based mobility as $\nu_{en} = e/(m_e\mu_{en})$, while the electron–ion collision frequency is calculated using the NRL Plasma Formulary as $\nu_{ei} = 2.91\times10^{-12} n_e[\mathrm{m}^{-3}]\ln\Lambda\, T_e[\mathrm{eV}]^{-3/2}\ \mathrm{s}^{-1}$. In the weakly ionized limit, where electron–neutral collisions dominate, the electrical conductivity is $\sigma_{\mathrm{en}} = n_e e^2/(m_e\nu_{en})$, whereas for a fully ionized plasma the electron–ion contribution can be written as $\sigma_{\mathrm{ei}} = 1.975\, n_e e^2/(m_e\nu_{ei})$ [26]. Combining these expressions gives the effective electron mobility

$$\mu_e = \frac{e}{m_e\left(\nu_{en} + \nu_{ei}/1.975\right)}. \tag{4}$$

Thus, the BOLSIG+-based electron mobility is corrected by the factor $\nu_{en}/(\nu_{en} + \nu_{ei}/1.975)$, yielding the effective mobility used in the model. The same correction factor is applied to the electron diffusion coefficient, D_e, to account for the additional momentum loss associated with electron–ion Coulomb collisions. The resulting electrical conductivity is compared with the LTE conductivity in Fig. fill later.

The ion–neutral momentum-transfer cross section is obtained from the fitted expression $\overline{Q}_{ia} = 2[7\times10^{-10} - 0.6\times10^{-10}\ln(2k_BT_h)]$ [28]. The low-field ion mobility is then calculated using the Mason–Schamp equation [33],

$$\mu_i = \frac{3e}{16N}\sqrt{\frac{2\pi}{\mu k_B T_{\mathrm{gas}}}}\,\frac{1}{\Omega^{(1,1)}(T_{\mathrm{gas}})}, \tag{5}$$

where N is the neutral number density, μ is the reduced mass of the ion–neutral pair, and $\Omega^{(1,1)}$ is the corresponding collision integral. This expression gives the ion mobility in the low-field limit, where the ion drift velocity is proportional to the applied electric field.

To account for the field dependence of the ion mobility, a closed-form two-temperature Wannier formulation is employed. The drift velocity is expressed as $v_d = \mu(T_{\mathrm{eff}})E$, where

$$\mu(T_{\mathrm{eff}}) = \mu_0\sqrt{\frac{T_{\mathrm{gas}}}{T_{\mathrm{eff}}}}, \qquad k_BT_{\mathrm{eff}} = k_BT_{\mathrm{gas}} + \frac{1}{3}Mv_d^2,$$

with μ_0 denoting the low-field mobility and M the ion mass. Substitution of these relations yields a quadratic equation in v_d^2, which is solved to obtain the field-dependent drift velocity. In the

limit $E \to 0$, the effective temperature approaches the gas temperature, $T_{\text{eff}} \to T_{\text{gas}}$, and the formulation recovers the low-field mobility, $\mu \to \mu_0$.

3.1 Quasi-Maxwellian electrons near the electrode

Even where the heavy species are strongly out of equilibrium, the electron population can remain closely Maxwellian provided electron–electron and electron–ion collisions are frequent enough to redistribute directed (drift) energy into the isotropic thermal spread faster than electron–heavy collisions or the imposed field can distort it. Following Dinulescu and Pfender's analysis of the anode boundary layer [29], two conditions must hold. First, electron–electron/ion thermalization must dominate electron–neutral momentum transfer, weighted by the mass ratio that makes each electron–neutral collision inefficient at removing energy:

$$\sum_h \frac{m_e}{m_h} \frac{\bar{\nu}_{eh}}{\bar{\nu}_{ee}} \ll 1. \tag{6}$$

Second, the ordered drift energy imparted by the field must remain small compared with the electron thermal energy, weighted by the same mass ratio:

$$\left(\frac{U_e}{\bar{c}_e}\right)^2 \ll \frac{m_e}{m_h}. \tag{7}$$

For a representative near-anode condition, $T_e \sim 1$ eV, $n_e \sim 5 \times 10^{22}\,\mathrm{m}^{-3}$, neutral density $N \sim 6 \times 10^{23}\,\mathrm{m}^{-3}$, electron–atom momentum-transfer cross section $\bar{Q}_{ea} = 2 \times 10^{-20}\,\mathrm{m}^2$, and mean thermal speed $\bar{c}_e = 7 \times 10^5\,\mathrm{m/s}$: the electron–ion/electron–electron frequency from the NRL formulary (with $\ln\Lambda \approx 3$) is $\nu_{ee} \sim \nu_{ei} \sim 5 \times 10^{11}\,\mathrm{s}^{-1}$, while the electron–neutral frequency is $\nu_{en} = \bar{c}_e N \bar{Q}_{ea} \approx 8 \times 10^9\,\mathrm{s}^{-1}$, i.e. $\nu_{en}/\nu_{ee} \sim 0.016$; with $m_e/M_{\mathrm{Ar}} \approx 1.4 \times 10^{-5}$, condition (6) gives $\sim 2 \times 10^{-7} \ll 1$. For condition (7), the electrical conductivity of the layer, $\sigma \approx 5000$ S/m, corresponds to an electron mobility $\mu_e = \sigma/(n_e e) \approx 0.6\,\mathrm{m}^2/(\mathrm{V\,s})$; with a representative current density $J \sim 10^7\,\mathrm{A/m}^2$, the electron drift speed is $U_e = J/(en_e) \approx 1.3 \times 10^3\,\mathrm{m/s}$, so $(U_e/\bar{c}_e)^2 \approx 3 \times 10^{-6}$, again well below $m_e/M_{\mathrm{Ar}} \approx 1.4 \times 10^{-5}$. Both conditions are comfortably satisfied, so the electron population is treated as quasi-Maxwellian throughout the near-electrode layer, and a single electron temperature T_e (rather than a fully kinetic electron energy distribution) is retained in equation (3) there as well as in the bulk.

3.2 Length scales and the thickness of the near-electrode layer

The same parameters fix a hierarchy of length scales that separates the space-charge sheath from the broader near-electrode non-equilibrium region that must be resolved around it. The electron Debye length,

$$\lambda_{De} = \sqrt{\frac{\varepsilon_0 T_e}{n_e e}} \approx 3 \times 10^{-8}\ \mathrm{m}, \tag{8}$$

sets the scale of the space-charge sheath itself. The electron mean free path against the combined electron–neutral and electron–electron/ion collision frequency used above,

$$\lambda_e = \frac{\bar{c}_e}{\nu_{en} + \nu_{ee}} \approx 1 \times 10^{-6}\ \mathrm{m}, \tag{9}$$

is ~ 30 times λ_{De}, confirming that the sheath is thin compared with a mean free path and can support the non-collisional electron kinetics discussed in Section 3.3. The electron–heavy energy-relaxation length, over which an electron transfers its excess thermal energy to the heavy gas through the inefficient mass-ratio-weighted elastic channel in equation (3), is larger still,

$$\lambda_T = \frac{\lambda_e}{\sqrt{2m_e/M}} \approx 2 \times 10^{-4}\ \mathrm{m}. \tag{10}$$

Finally, the diffusion length that controls how far the electron/ion density profile itself can depart from its local equilibrium value under a given field, used directly by Lowke and Tanaka's mesh-sizing rule [10], is $\Lambda = (kT_e/e)/E$; with $T_e \approx 1$ V and a field $E = J/\sigma \approx (10^7\,\mathrm{A/m}^2)/(5000\,\mathrm{S/m}) = 2000$ V/m consistent with the parameters above,

$$\Lambda \approx 5 \times 10^{-4}\ \mathrm{m}. \tag{11}$$

The four scales satisfy $\lambda_{De} \ll \lambda_e \ll \Lambda \sim \lambda_T$, the same ordering (Debye length $\ll$ mean free path $\ll$ boundary-layer thickness) reported directly in Table I of [29] for the anode boundary layer of a

comparable arc, $\lambda_{De} \sim (2\text{–}3) \times 10^{-8}$ m, $\lambda_e \sim (1\text{–}2) \times 10^{-6}$ m, boundary-layer thickness $L \sim 0.2$–0.6 mm. On this basis we take a near-electrode layer of thickness $\delta \approx 0.2$ mm — comparable to both Λ and λ_T, and consistent with the range of L found in [29] — as the region over which the unified, non-equilibrium model of Section 4 must be applied; outside it, the plasma is close enough to collision-dominated equilibrium for the LTE bulk treatment of Section to hold. Note that this ∼0.2 mm layer is not itself the space-charge sheath: it is the outer, thermal/diffusion-dominated non-equilibrium region (corresponding to layers 2–3 of the four-sublayer structure found by [18]) inside which a much thinner space-charge sheath, of order λ_{De} to a few times λ_{De}, is nested.

3.3 Is the sheath collisionless?

A collisionless sheath is the standard simplifying assumption used to close the NLTE-sheath and two-temperature ionization-layer-sheath approaches of Section 1.1, and it is tempting to expect it to hold automatically at atmospheric pressure, where collision rates are high. Almeida, Benilov and Naidis show this expectation is wrong: computing the number of ion collisions actually incurred while crossing the sheath gives values from order unity to 10^4, not $\ll 1$, even for atmospheric-pressure argon [18]. The reason is that the sheath thickness itself scales with the *local* Debye length, $\lambda_{De} \propto n_e^{-1/2}$, which is set by the arc's own high electron density and shrinks faster, as the arc becomes hotter and more strongly ionized, than the neutral mean free paths shrink with pressure (the Ramsauer minimum in the argon electron–neutral cross section keeps λ_e from falling as quickly as λ_{De}). A collisionless sheath at high pressure is therefore not evidence of a crude or incomplete model; it is close to the generic outcome for a dense atmospheric-pressure arc, and the question that actually needs answering species by species and quantity by quantity is *how* collisionless, and for what.

We make this quantitative with four Knudsen numbers, each the ratio of the relevant mean free path to the resolved sheath width s: $\mathrm{Kn}_e = \lambda_e/s$ (do electrons collide crossing the sheath at all); $\mathrm{Kn}_\varepsilon = \lambda_T/s$ (electron–heavy energy exchange: does a local electron energy balance exist inside the sheath); $\mathrm{Kn}_{ee} = \lambda_{ee}/s$ (electron–electron thermalization: does a well-defined T_e, i.e. a locally Maxwellian electron energy distribution, exist inside the sheath at all – a distinct and much more restrictive question than Kn_ε, since λ_{ee} and λ_T differ by roughly 10^4); and $\mathrm{Kn}_i = \lambda_i/s$ for the ion charge-exchange mean free path. For the converged, resolved-sheath operating point of the PlasmeX test case discussed in Section 4 ($s = 1.65 \times 10^{-7}$ m $= 5.6\,\lambda_{De}$, $T_e \approx 0.98$ eV at the wall), these evaluate to $\mathrm{Kn}_e \approx 63$, $\mathrm{Kn}_\varepsilon \approx 1.2 \times 10^4$, $\mathrm{Kn}_{ee} \approx 5.3$, and $\mathrm{Kn}_i \approx 0.8$–$1.7$: the sheath is collisionless for electron momentum and electron–heavy energy exchange by a wide margin, only marginally collisionless for electron thermalization, and carries of order one ion collision, consistent with the lower end of the 10^2–10^4 range quoted by [18] for a somewhat different case. Despite $\mathrm{Kn}_{ee} \sim 5$ being only marginal, an electron population entering a purely retarding, elastic potential barrier is mapped onto itself by Liouville's theorem: reflection by an energy-conserving barrier preserves a Maxwellian at unchanged T_e while depleting the density by the Boltzmann factor, giving a testable, parameter-free reference prediction, $T_e(x) = T_{e,\mathrm{edge}} = \text{const}$ across the sheath, against which the local closure used in equation (3) can be checked directly (Section 4). No such collisionless reference state exists for the ions, which do not see a retarding barrier and for which $\mathrm{Kn}_i = O(1)$ rules out both the collisionless limit and full collisional equilibrium; ion behaviour in the sheath is therefore treated as an explicit target for the resolved drift-diffusion solution rather than assumed a priori in either limit.

4 Unified near-electrode model and numerical implementation

4.1 Resolving the sheath directly, rather than matching or homogenizing it

Section 1.1 distinguished three published strategies for coupling a bulk-plasma closure to the electrode: solving one boundary value problem with Poisson's equation over the whole gap [18]; matching a non-equilibrium bulk solution directly to an analytic sheath closure at an interface, valid strictly only for a collision-dominated sheath [20]; and embedding the unresolved sheath as an effective electrical conductivity in the boundary-adjacent finite-volume cells of an otherwise quasi-neutral CFD solution, $\sigma_{\mathrm{eff}} = [(1 - I_g)/\sigma_{\mathrm{child}} + I_g/\sigma]^{-1}$ with σ_{child} a Child-law space-charge-limited conductivity [25]. We use a fourth strategy, implemented in the PlasmeX code: solve the same drift-diffusion– Poisson system, equations (1)–(2), in the bulk plasma and through the sheath alike, and use block-structured adaptive mesh refinement (AMR), built on the AMReX framework, to refine the mesh at the electrode down to a fraction of the local Debye length. No separate sheath closure, matching condition, or homogenized conductivity is introduced; the degree of local non-equilibrium is a property of the converged solution, not a modelling choice made in advance.

This is closest in spirit to the unified modelling of [18], generalized from a specialized one-dimensional boundary-value solver to a general-purpose, AMR finite-volume PDE code, and it should be contrasted directly with the homogenized effective-conductivity shortcut of [25], which is the more practical choice for genuinely three-dimensional geometries where literal sub-Debye-length mesh refinement is not affordable.

4.2 Transport closures

Electron mobility and diffusion coefficients are obtained from a pre-tabulated, effectively Maxwellian two-term Boltzmann solution, $\mu_e N(T_e)$ and $D_e N(T_e)$. This is verified, not assumed: across the full tabulated range the ratio $(D_e/\mu_e)/T_e$ deviates from unity by 0.10% rms (worst case 0.50%), which is exactly the Einstein relation for a Maxwellian electron energy distribution, and confirms that at the 0.9–10% ionization fractions relevant here (ν_{ei}/ν_{en} ranging from 12 to 143 over $T_{\mathrm{gas}} = 1000$–$12\,000$ K) electron–electron and electron–ion collisions dominate every inelastic channel strongly enough to Maxwellize the bulk electron population, exactly as required by Section 3.1. The electron–ion Coulomb correction, equation (**??**), reduces the neutral-collision-only conductivity by one to three orders of magnitude and corrects its qualitative trend (neutral-only conductivity rises with temperature only because it divides by a falling neutral density; the corrected, Spitzer-like conductivity, $\propto T_e^{3/2}/\ln\Lambda$, falls in the $\sim 10^3$ S/m range expected for LTE argon at 1 atm). A Saha-equilibrium cross-check on the arc-core initial condition, $n_e^{\mathrm{Saha}} = 9.909 \times 10^{22}\,\mathrm{m}^{-3}$ against the solved $n_e = 9.860 \times 10^{22}\,\mathrm{m}^{-3}$, agrees to 0.5%, confirming the bulk solution is a genuine LTE state without that equilibrium being imposed. Ion transport uses the closed-form two-temperature Wannier closure of Section 2.1; at the reduced field found at the converged electrode cell of the sheath test case below (5976 Td), it gives $\mu/\mu_0 = 0.16$, a factor-of-six reduction from the low-field mobility, and recovers $\mu \to \mu_0$ in the low-field bulk.

4.3 Electrode boundary conditions

At the electrode face, the electron flux is set to the standard kinetic wall flux $\Gamma_e = \frac{1}{4} n_e \bar{c}_e$ evaluated on the (self-consistently depleted) wall density, and the ion flux uses a Hagelaar-type drift-aware closure, $\Gamma_i = \frac{1}{4} n_i \bar{c}_i + \max(\mu_i n_i E, 0)$, that reduces to the thermal flux at zero field and adds the outward drift contribution when the local field drives ions into the wall. This is structurally the same closure used for the counter-diffusing current in the NLTE-sheath boundary condition of [20] (their J_{CD}, equation A.7) and for the back-diffusion current in [25] (their J_{bd}, equation 11): a Boltzmann-retarded thermal flux evaluated at the wall density and local T_e. Two implementation choices in the present model depart from a fully symmetric, field-aware treatment of both species and are noted here as open items (Section 5 / `todo.txt`): the electron wall flux is currently evaluated with the field-independent $\frac{1}{4} n_e \bar{c}_e$ form rather than a Hagelaar-type field-aware form for electrons as well, and no secondary or thermionic electron emission term is included at the electrode, which is normally the dominant electron supply at a DC arc cathode and would need to be added before any cathode-fall prediction from this model could be considered complete.

4.4 Numerical implementation

The model is implemented in PlasmeX, a C++17 drift-diffusion plasma code built on the AMReX block-structured AMR framework, currently configured for a three-species (e^-, Ar^+, Ar) argon plasma. The electrostatic potential is solved with AMReX's geometric multigrid solver (MLMG) in general, and with Hypre's structured multigrid solver (`HYPRE_StructPFMG`) for the anisotropic two-dimensional axisymmetric mesh used in production runs, since MLMG's composite multigrid cycle was found not to converge on that mesh at any tested aspect ratio, bottom solver, or coarsening depth. Species transport uses a semi-implicit, operator-split scheme: drift fluxes are upwinded explicitly and diffusive fluxes are backward-Euler in an implicit face-coefficient form, $1 - \Delta t\,(e/\varepsilon_0)\mu_e n_e$, that removes the explicit dielectric-relaxation time-step restriction of a fully explicit electron drift treatment and gives roughly a 100× larger stable time step at equal accuracy, verified by matching both schemes at matched physical time to $\sim 10^{-6}$ relative agreement after a long integration from a common checkpoint.

In one spatial dimension, both linear solves (the electrostatic Poisson solve and the implicit species-diffusion solve) reduce to a tridiagonal system on each AMR level, for which a direct solver is both faster and more accurate than the general-purpose iterative multigrid solver: instrumented at 93% of a time step for a 512-cell problem, the iterative solver is replaced by a direct Thomas-algorithm solve whose matrix entries are recovered once, exactly, by probing the discrete operator, giving an 8.9× reduction in total time-step cost (65× and 13× on the potential and species solves individually) with an exact rather than iteratively converged linear-algebra answer.

This extends, non-trivially, to the full composite AMR hierarchy: refluxing at a coarse-fine interface exactly cancels the covered coarse cell and replaces it with the fine-level flux, and the coarse-to-fine ghost-cell interpolation used in one dimension is an exact two-term closure over a single coarse cell, ghost $= \frac{2}{3}x_C + \frac{1}{3}x_{f_0}$ at a refinement ratio of 2 — together these guarantee that the composite operator over the leaf cells of a refined 1D hierarchy is itself exactly tridiagonal, so the same direct solve applies unmodified across level boundaries. This direct solver is what makes literal AMR refinement to sub-Debye-length cells in the sheath computationally practical rather than prohibitively expensive; it is restricted to one spatial dimension and a single MPI rank, so the production two-dimensional axisymmetric configuration falls back to Hypre as noted above.

5 Results

6 Conclusion

Acknowledgments

To be completed.

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